Vivek Jha

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EDUCATION

Indian Institute Of Information Technology Bhopal — Bhopal, Madhya Pradesh , India
Bachelor Of Technology in Information Technology; CGPA: 8.48 — *Aug. 2024 – April. 2028*

Alpha Junior College Of Science and Commerce — Mumbai , India
Grade XII; Percentage: 90.33 — *Aug. 2022 – April. 2024*

Shree Chandulal Nanavati Vinaymandir School — Mumbai, India
Grade X; Percentage: 95.8 — *June. 2021 – April. 2022*

PROJECTS

- **Abuse-Ring Sentinel** (GitHub): **Graph-Based Fraud Intelligence & Response Platform** | **Tech Stack:** Python, FastAPI, SQL, Graph Analysis. Built a fraud-intelligence platform using point-in-time-safe temporal features and strict time-based train/validation/held-out splits. Added leakage-prevention tests and graph-based investigation to surface suspicious entities and potential abuse rings.

- **CertiMail** (GitHub): **Certificate Distribution & Personalized Email Automation** | **Tech Stack:** Python, FastAPI, React, TypeScript, SQLite, SMTP. Built a full-stack tool for spreadsheet-driven certificate matching and personalized email campaigns, with mismatch review and approval, template previews, test sends, CC/BCC, delivery logs, and recipient-level tracking.

- **Acoustic Environment Classifier** (GitHub): Resume Project Title Acoustic Environment Classification of YouTube Audio using Deep Neural Networks **Tech Stack:** Python, TensorFlow, Keras, Librosa, Scikit-Learn, PCA, SHAP, AudioSet, PANNs, YAMNet, VGGish. Developed a deep learning model to classify audio as Urban or Natural using 13,000+ clips from Google AudioSet. Built an end-to-end audio processing pipeline with feature extraction and transfer learning using PANNs, YAMNet, and VGGish embeddings. Applied PCA and trained an MLP in TensorFlow/Keras, achieving 90.1% accuracy and 0.766 Test AUC. Used SHAP to interpret model predictions and identify key acoustic features influencing classification.

- **Tamil Nadu Traffic Rules Assistant (RAG Chatbot)** (GitHub): **AI-Powered Traffic Rules Assistant using Retrieval-Augmented Generation (RAG) Tech Stack:** Python, LangChain, FAISS, Sentence Transformers, FastAPI, Groq LLM, NLP, RAG, Vector Search. Built a RAG-based chatbot to answer queries on Tamil Nadu traffic regulations using government documents. Implemented PDF processing, semantic chunking, embeddings, and FAISS vector search for accurate document retrieval. Integrated LangChain with Groq LLMs to generate context-aware, grounded responses with reduced hallucinations. Developed REST APIs using FastAPI and optimized retrieval for low-latency, explainable responses.

PROGRAMMING SKILLS

- **Languages:** Python, Javascript, C++, SQL, Java
- **Technologies:** AWS, Scikit-learn, React, Kafka, Keras, Github, Docker, Langchain, Tensorflow, Pytorch, FastApi, Django, Flask

EXTRA-CURRICULARS

- **Chairperson, IEEE Student Branch, IIIT Bhopal**
  Lead the student branch by coordinating technical events, workshops, industry collaborations, sponsorships, and a cross-functional executive team to foster professional and technical development among students.
- **Outreach Executive, CODAME – Competitive Programming Club, IIIT Bhopal**
  Managed outreach and promotional activities, coordinated with partner organizations and student communities, and contributed to increasing participation and engagement in competitive programming contests and coding events.
- **Published Poet, Shankh College Magazine**
  Authored and published original poetry in the college's annual magazine, showcasing creativity, effective written communication, and literary expression.